

Triangles (Set B) — MCQ Worksheet

Mathematics • Chapter 6 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Two figures having same shape (and not necessarily same size) are called:

- (A) Congruent
- (B) Similar
- (C) Equal
- (D) Parallel

Q2. In $\triangle ABC$, $DE \parallel BC$ and D, E are on AB, AC respectively. If $AD/DB = 3/4$, then $AE/EC =$

- (A) $3/4$
- (B) $4/3$
- (C) $1/2$
- (D) $3/7$

Q3. If $\triangle ABC \sim \triangle PQR$ and $AB/PQ = 1/3$, then ratio of areas (ABC : PQR) =

- (A) 1:3
- (B) 1:6
- (C) 1:9
- (D) $1:\sqrt{3}$

Q4. By AA similarity, $\triangle ABC \sim \triangle DEF$ if:

- (A) Two angles of one equal two angles of other
- (B) All sides equal
- (C) Hypotenuses equal
- (D) Two sides proportional

Q5. If perimeters of two similar triangles are in ratio 4:5, ratio of their corresponding sides is:

- (A) 4:5
- (B) 16:25
- (C) $2:\sqrt{5}$
- (D) 5:4

Q6. In right $\triangle ABC$, right-angled at B, $AB = 5$ and $BC = 12$. $AC =$

- (A) 13
- (B) 17
- (C) 169
- (D) 8.5

Q7. Triangles with sides 3, 4, 5 and 6, 8, 10 are:

- (A) Similar but not congruent
- (B) Congruent
- (C) Neither
- (D) Cannot say

Q8. If a line is drawn parallel to one side of a triangle to intersect the other two sides, it divides the two sides in the:

- (A) Same ratio
- (B) Ratio 1:2
- (C) Equal halves
- (D) Ratio of the third side

Q9. Pythagoras' theorem states that in a right triangle:

- (A) Sum of squares of legs = square of hypotenuse
- (B) Sum of legs = hypotenuse
- (C) Square of leg = product of legs
- (D) $\text{Hypotenuse}^2 = \text{leg}$

Q10. In $\triangle ABC$, $\angle B = 90^\circ$. If $AB = 9$ cm and $AC = 15$ cm, then $BC =$

- (A) 12 cm
- (B) 9 cm
- (C) 15 cm
- (D) 18 cm

Q11. Two similar triangles have areas in ratio 25:49. The ratio of their perimeters is:

- (A) 5:7
- (B) 25:49
- (C) 7:5
- (D) $\sqrt{5}:\sqrt{7}$

Q12. If $\triangle PQR \sim \triangle XYZ$ and $\angle P = 50^\circ$, $\angle Q = 60^\circ$, then $\angle Z =$

- (A) 50°
- (B) 60°
- (C) 70°
- (D) 90°

Q13. In $\triangle ABC$, AD is median. Then $AB^2 + AC^2 =$

- (A) $2(AD^2 + BD^2)$
- (B) $AD^2 + BD^2$
- (C) $AD^2 - BD^2$
- (D) $BD^2 - AD^2$

Q14. All squares are:

- (A) Similar
- (B) Congruent
- (C) Both similar and congruent only if equal sides
- (D) Neither

- Q15.** If $\triangle ABC \sim \triangle DEF$, $\text{area}(ABC) = 64 \text{ cm}^2$ and $\text{area}(DEF) = 121 \text{ cm}^2$, and $BC = 15.4 \text{ cm}$, then $EF =$
 (A) 11.2 cm (B) 21.2 cm
 (C) 15.4 cm (D) 13 cm
- Q16.** In $\triangle ABC$, P and Q are points on AB and AC respectively such that $PQ \parallel BC$. If $AP = 4 \text{ cm}$, $PB = 6 \text{ cm}$ and $AQ = 5 \text{ cm}$, then $AC =$
 (A) 12.5 cm (B) 7.5 cm
 (C) 15 cm (D) 10 cm
- Q17.** If sides of two similar triangles are in ratio 4:9, area of larger triangle is 81 m^2 . Area of smaller is:
 (A) 16 m^2 (B) 24 m^2
 (C) 36 m^2 (D) 9 m^2
- Q18.** An aeroplane leaves an airport and flies due north at 1000 km/h . At the same time another flies due west at 1200 km/h . After $1\frac{1}{2} \text{ h}$, the distance between the two is:
 (A) $1500\sqrt{(1+1.44)} \text{ km} \approx 2342 \text{ km}$ (B) $1500\sqrt{(1^2+1.2^2)} \text{ km} = 1500\sqrt{2.44} \text{ km}$
 (C) About 2343 km (D) Both A and C
- Q19.** In $\triangle ABC$, $AB^2 + AC^2 = 5 BC^2$. If AD is perpendicular from A to BC, then $BD/DC =$
 (A) 2:3 (B) 3:2
 (C) 1:4 (D) 4:1
- Q20.** Two poles of heights 6 m and 11 m stand on a plane ground. If the distance between their feet is 12 m, distance between their tops is:
 (A) 13 m (B) 12 m
 (C) 15 m (D) 17 m
- Q21.** A ladder 25 m long reaches a window 24 m above the ground. The distance of foot of ladder from the wall is:
 (A) 7 m (B) 8 m
 (C) 15 m (D) 17 m
- Q22.** In $\triangle ABC$, D is midpoint of BC and $AD \perp BC$. Then $AB^2 + AC^2 =$
 (A) $2(AD^2 + BD^2)$ (B) $2AD^2 + BD^2$
 (C) $AD^2 + 2BD^2$ (D) $AD^2 - BD^2$
- Q23.** In a triangle ABC right-angled at C, if D is the mid-point of BC, then $AB^2 =$
 (A) $4AD^2 - 3AC^2$ (B) $AD^2 + AC^2$
 (C) $AD^2 - AC^2$ (D) $3AD^2 - 4AC^2$
- Q24.** In $\triangle ABC$, $\angle A = 90^\circ$ and $AD \perp BC$. If $BD = 4 \text{ cm}$ and $CD = 9 \text{ cm}$, then $AD =$
 (A) 6 cm (B) 5 cm
 (C) 6.5 cm (D) 13 cm
- Q25.** The diagonals of a quadrilateral ABCD intersect at O. If $OA \times OC = OB \times OD$, then quadrilateral is a:
 (A) Trapezium (B) Rhombus
 (C) Square (D) Cyclic